The Analyst Recommendation and Earnings Forecast Anomaly

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Equity Anomalies Extra Credit Project 17

Point-in-time IBES signals, CRSP returns, and a leakage-resistant backtest

Research Design Presentation

Cost-sensitivity results available; full performance audit in progress

Research question, mechanism, and hypotheses

Research question

Can public analyst recommendations predict *subsequent* stock returns after conservative information lags and trading costs?

Economic mechanism

- Analysts process firm, industry, and management information.
- Recommendation news may be incorporated gradually.
- Revisions may isolate new information better than recommendation levels.

Pre-specified hypotheses

1. Favorable consensus predicts higher next-month returns.
2. Upgrades outperform downgrades.
3. Signals strengthen when aligned with 12–2 momentum.
4. Effects may be larger for low-coverage firms.

Identification target

Return predictability must arise **after** the IBES information becomes observable—not from same-month returns, current identifiers, or missing outcome selection.

Data, historical links, and signal construction

Licensed data inputs

- **IBES Summary:** ticker, CUSIP, `statpers`, `meanrec`, `numrec`.
- **CRSP monthly:** PERMNO, return, delisting return, price, shares, share/exchange codes.
- **CRSP names:** historical CUSIP intervals for point-in-time linking.

Local Parquet cache

One-time WRDS download; files are excluded from Git and reused unless `-force` is supplied.

Consensus level

$$s^{C}_{i,t} = -\,\mathrm{meanrec}_{i,t}$$

Lower IBES scores are more favorable, so the sign is reversed.

Recommendation revision

$$s^{R}_{i,t} = \mathrm{meanrec}_{i,t-1} - \mathrm{meanrec}_{i,t}$$

Positive values are upgrades; negative values are downgrades.

Planned conditioning variables: 12–2 CRSP

Point-in-time design and return alignment



Availability rule

Use the latest signal satisfying

$$\text{available_date}_i \leq \text{formation_date}_t.$$

A month-end snapshot cannot be traded at that same close.

Outcome rule

Weights formed in month t join only to calendar-month $t+1$ returns. CRSP outcomes combine

$$(1 + r_{i,t+1})(1 + \mathit{dlret}_{i,t+1}) - 1.$$

Portfolio construction and performance measurement

1. Restrict to CRSP common stocks on NYSE/AMEX/NASDAQ.
2. Require price $\geq$ \$5 and at least two analysts.
3. Rank the chosen signal within each formation month.
4. Long the top decile; short the bottom decile.
5. Baseline legs are equal weighted with gross exposure 200%.
6. Apply month-t weights to month-$t+1$ total returns.

Net strategy return

$$r_{p,t+1}^{net} = \sum_i w_{i,t} r_{i,t+1} - \frac{c}{10{,}000} \sum_i |w_{i,t} - w_{i,t-1}|,$$

Primary specifications

Signal	Weight	Holding
Consensus	Equal	1 month
Consensus	Value	1 month
Revision	Equal	1 month
Revision	Value	1 month
Revision $\times$ momentum	Equal	1 month

Report

- CAGR and annualized mean/volatility
- Sharpe ratio and maximum drawdown
- Turnover, costs, counts, and coverage
- Factor alpha with robust inference

Leakage controls, automated tests, and data audit

Controls embedded in the pipeline

- Historical CUSIP validity; never current-ticker matching.
- Signal ranking does not inspect future-return availability.
- Missing held outcomes invalidate the portfolio month; never set to zero.
- Complete entries, exits, and rebalancing count toward turnover.
- Duplicate PERMNO/snapshot rows fail before revision construction.
- Extra CRSP outcome month cannot become a new formation month.

Seven automated tests currently pass

- Signal unavailable before publication date.
- Month-end snapshot receives the one-day lag.
- Future revisions cannot alter past signals.
- Month t maps to calendar month $t + 1$.
- Delisting returns compound correctly.
- Missing held outcome is not treated as zero.
- Revision uses only the prior snapshot.

Real-data gate

Before interpreting returns: audit link rate, ambiguous identifiers, duplicates, date coverage, delistings, outcome coverage, and portfolio counts.

Feasible extensions and required WRDS products

Extension	Additional design	Data	Feasibility
Momentum interaction	Double-sort revisions with trailing 12–2 returns	Existing CRSP	High
Coverage anomaly	Split signal efficacy by `numrec`	Existing IBES	High
Disagreement	Recommendation dispersion or analyst-detail cross-section	IBES optional/detail	Medium
Fundamentals	E/P, book-to-market, profitability, accruals	Compustat/ratios	Medium
Forecast revisions	Price-scaled changes in expected EPS	IBES estimates	Medium
PEAD interaction	Earnings surprise $\times$ recommendation revision	IBES actuals + CRSP	Medium–hard
Analyst skill	Condition on prior forecast/recommendation performance	IBES detail IDs	Hard
Broker quality	Point-in-time broker identity and classification	IBES detail + map	Hard

Recommended scope

Complete consensus and revision baselines first; then add momentum and coverage. Delay analyst/broker-level work because IBES identifiers have been reassigned historically.

Empirical result: transaction costs materially weaken performance

Sharpe ratio by one-way cost assumption

Cost	Consensus	Revision
0 bps	0.071	**0.513**
2 bps	0.059	0.373
5 bps	0.041	0.164
10 bps	0.011	−0.186

Consensus signal

Sharpe is low even before costs (0.071), but it decays relatively gradually to 0.011 at 10 bps.

Revision signal

Gross Sharpe is much stronger (0.513), yet the signal is highly cost-sensitive and becomes negative at 10 bps.

Economic implication

Revision predictability is only potentially implementable with low turnover, low trading costs, or a longer holding/rebalancing design.

Main conclusion

The revision strategy dominates before costs, but its apparent advantage is not robust to conservative implementation costs. Turnover reduction is now a central research question, not a secondary adjustment.

Remaining results and robustness to complete

Strategy	Ann. mean	CAGR	Ann. vol.	Sharpe	Max DD	Turnover
Consensus EW	TBD	TBD	TBD	0.071*	TBD	TBD
Consensus VW	TBD	TBD	TBD	TBD	TBD	TBD
Revision EW	TBD	TBD	TBD	0.513*	TBD	TBD
Revision VW	TBD	TBD	TBD	TBD	TBD	TBD
Revision × momentum	TBD	TBD	TBD	TBD	TBD	TBD

*Gross Sharpe at 0 bps; net-cost sensitivity is reported on the preceding slide.

Inference

- Monthly mean with HAC/Newey–West standard errors
- CAPM and multi-factor alpha
- Economic effect after transaction costs

Robustness

- Alternative lags, holding periods, and quantiles
- Size, price, liquidity, and coverage screens
- Subperiods and equal/value weighting

Interpretation rule

Do not claim an anomaly from one favorable specification. Emphasize stability, implementation costs, factor adjustment, and the number of alternatives tested.

Limitations, execution plan, and primary sources

Known limitations

- WRDS schemas and entitlements differ across institutions.
- IBES analyst/broker identifiers have changed over time.
- Exact intraday publication times are generally unknown.
- Equal weighting can exaggerate small-stock implementability.
- Multiple specifications create data-mining risk.

Primary sources

- WRDS I/B/E/S Overview and ICLINK documentation
- WRDS CRSP Monthly Stock documentation
- *Handbook of Equity Market Anomalies*, Chapter 3

Execution plan

1. Complete one-time WRDS Parquet cache.
2. Audit schemas, links, dates, duplicates, and outcomes.
3. Run consensus and revision EW/VW baselines.
4. Add momentum, coverage, costs, and factor regressions.
5. Replace the two reserved frames with audited results.

Reproducible artifacts

`src/` extraction, signals, and backtest
`tests/` leakage checks `notebooks/` workflow